

captures how much experience an occupation *requires* rather than the actual seniority of its workforce, providing an independent check that the Revelio-based classification aligns with established occupational characteristics.

O*NET task-to-occupation mapping. The AEI matches conversations to O*NET task descriptions, and the O*NET task statements database provides the mapping from tasks to detailed occupations (8-digit O*NET-SOC codes). Each task may appear in multiple occupations, and each occupation contains multiple tasks. We merge the AEI usage data with this mapping to create a dataset at the task–occupation level, where each observation represents a specific O*NET task within a specific occupation, with associated GenAI usage statistics. After merging, the analysis sample contains 4,494 task–occupation pairs spanning 753 occupations across 22 two-digit SOC groups.

A.7.2 Descriptive Patterns

Table A.1 reports average GenAI usage patterns by Revelio seniority tercile. To summarize the balance between automation and augmentation in GenAI usage, we construct two measures at the task level: *displacement intensity*, defined as the share of classified GenAI interactions (displacement + augmentation) accounted for by displacement-type modes (directive and feedback loop), and *augmentation intensity*, defined analogously for augmentation-type modes (learning, validation, and task iteration). The two sum to 100% by construction for each task.

The aggregation from raw conversations to the tercile-level statistics in Table A.1 proceeds as follows. First, the AEI assigns each conversation a collaboration mode and maps it to an O*NET task; we compute mode shares at the task–occupation level (the share of conversations for a given task in a given occupation that fall into each mode). Second, we aggregate to the occupation level by taking weighted averages of task-level mode shares, with weights proportional to AEI task usage counts. Third, we classify occupations into terciles based on Revelio seniority and report unweighted averages across occupations within each tercile. The regression analyses in Table A.2 use the occupation-level measures as dependent variables. The sample shrinks from 753 occupations (full merged sample) to 559 in Table A.1 (requiring ≥ 50 Revelio workers for tercile assignment) and to 387 in Table A.2 (additionally requiring ≥ 3 AI-used tasks and non-missing seniority).

The central finding is that displacement intensity exceeds 50% only for junior-type occupations (52.0%), meaning that when GenAI is used for tasks characteristic of these occupations, the typical task tilts toward automation. For senior-type occupations, augmentation intensity dominates at 59.6% (displacement intensity: 40.4%). In other words, tasks in junior-intensive occupations are more likely to engage GenAI as a substitute, while tasks in senior-intensive occupations are more likely to engage GenAI as a complement.

This pattern is driven by the augmentation side. The individual collaboration mode shares show that task iteration rises monotonically with seniority (12.5% to 22.8%) and validation increases from 1.1% to 3.7%, while the directive share remains roughly flat across terciles. We also report the average O*NET experience requirements for each tercile as a complementary validation: occupations classified as junior-type by Revelio indeed require substantially less experience (1.4 years vs. 3.6 years for senior-type), confirming that the two seniority measures capture similar variation.

A.7.3 Regression Evidence

To assess whether the descriptive patterns hold within broad occupation groups, we regress occupation-level GenAI usage shares on the seniority measures, controlling for two-digit SOC fixed effects with heteroskedasticity-robust standard errors. This comparison exploits variation in seniority *within* occupation families—for instance, comparing secretaries to managers within “Office and Administrative Support,” or paralegals to lawyers within “Legal” occupations.

Table A.2 reports the results. The key finding is that displacement intensity *declines* significantly with seniority, confirming the descriptive pattern within narrow occupation groups. A one-standard-deviation increase in Revelio seniority is associated with a 3.68 percentage point decline in displacement intensity with 2-digit SOC fixed effects ($p = 0.002$). The individual mode shares tell a consistent story: the augmentation share increases significantly with seniority, driven primarily by task iteration (1.74 pp, $p = 0.029$) and validation (0.82 pp, $p = 0.005$), while the displacement share declines (−1.91 pp, $p = 0.018$). These results are consistent with the framework’s prediction that GenAI serves as a substitute for routine task execution in junior-intensive occupations and as a

Table A.1: AI Usage Patterns by Revelio Seniority Tercile

	Low (Junior-type)	Medium	High (Senior-type)
Number of occupations	151	202	206
Avg. Revelio seniority	1.36	2.03	3.11
Avg. experience (years, O*NET)	1.39	2.34	3.57
<i>Collaboration mode shares (%)</i>			
Directive	24.4	23.9	26.1
Feedback loop	3.4	2.2	3.0
Learning	13.0	16.7	15.5
Validation	1.1	2.6	3.7
Task iteration	12.5	17.9	22.8
Displacement intensity (%)	52.0	42.0	40.4
Augmentation intensity (%)	48.0	58.0	59.6

Notes: Each observation is an occupation. Occupations are divided into terciles based on the average seniority level of workers observed in Revelio Labs/LinkedIn data as of January 2023, restricted to occupations with ≥ 50 observed workers. Collaboration mode shares are weighted averages across an occupation's tasks, with weights proportional to AEI task usage counts. Displacement and augmentation intensities are computed at the task level as displacement/(displacement + augmentation) and augmentation/(displacement + augmentation), respectively, then averaged across tasks within each occupation and across occupations within each tercile. Displacement-type interactions include directive and feedback loop modes; augmentation-type interactions include learning, validation, and task iteration modes. The five collaboration mode shares do not sum to 100% because a substantial fraction of task-matched conversations are not confidently assigned to any mode by the AEI classification algorithm; displacement and augmentation intensities are computed using only the classified subset and sum to 100% by construction. Values above 50% for displacement intensity indicate that GenAI usage for a given task tilts toward automation rather than augmentation. The AEI data cover Claude interactions during November 13–20, 2025.

complement to expert judgment in senior-intensive occupations.

A.7.4 Discussion

The AEI analysis provides descriptive evidence consistent with the core mechanism of the conceptual framework. The framework assumes that GenAI is more likely to automate the routine, codifiable tasks that constitute the bulk of junior work, while the complex, judgment-intensive tasks that characterize senior work are more likely to involve augmentation-type GenAI usage. The data show that this asymmetry manifests

Table A.2: Revelio Seniority and GenAI Usage Mode: Occupation-Level Regressions

Dependent variable	No fixed effects			2-digit SOC FE		
	β	SE	p	β	SE	p
Directive	−1.00	0.74	0.18	−1.15	0.80	0.15
Learning	−0.18	0.71	0.80	0.22	0.89	0.81
Validation	0.83	0.26	0.002	0.82	0.29	0.005
Task iteration	2.55	0.71	<0.001	1.74	0.79	0.029
Displacement	−1.25	0.77	0.11	−1.91	0.80	0.018
Augmentation	3.20	0.92	<0.001	2.78	1.05	0.008
Displacement intensity	−3.26	1.02	0.002	−3.68	1.15	0.002

Notes: $N = 387$ occupations (with ≥ 3 AI-used tasks, non-missing seniority, and ≥ 50 Revelio workers). The independent variable is the Revelio seniority z-score (standardized average seniority from LinkedIn profiles, January 2023). Each row is a separate regression. Dependent variables are occupation-level weighted averages of GenAI collaboration mode shares, with weights proportional to task usage counts. Displacement = directive + feedback loop. Augmentation = learning + validation + task iteration. Displacement intensity = displacement/(displacement + augmentation) $\times 100$, computed at the task level and then averaged to the occupation level. Heteroskedasticity-robust standard errors.

not merely in which tasks are exposed to AI, but in *how* AI is used: junior-type tasks are engaged through directive interactions that substitute for human execution, while senior-type tasks are engaged through iterative and learning-oriented interactions that complement human expertise. This pattern—a qualitative difference in the mode of human–AI collaboration across the seniority spectrum—provides descriptive support for the aggregate employment patterns documented in the main analysis.

Several caveats apply. First, the AEI data classify conversations by the O*NET task they address, not by the seniority of the individual user. We observe how AI is used *for* tasks that are characteristic of junior or senior occupations, but we cannot determine whether a junior or senior worker initiated the interaction. The relevant variation is therefore at the task level, not the worker level. Second, the data cover a single week in November 2025 on a single AI platform, and usage patterns may evolve as the technology and its user base mature. Third, the collaboration mode classification is algorithmic and imperfect; some interactions may be misclassified. Despite these limitations, the consistency of the patterns across occupation-level and task-level specifications and across a range of professional domains suggests that the qualitative pattern is stable across specifications.